

Problem

A series RC circuit has $|V_R| = 12\text{ V}$ and $|V_C| = 5\text{ V}$. The magnitude of the supply voltage is:

- (a) -7 V
- (b) 7 V
- (c) 13 V
- (d) 17 V

Step-by-step solution

Step 1 of 2

Consider a simple series circuit as shown in Figure 1.

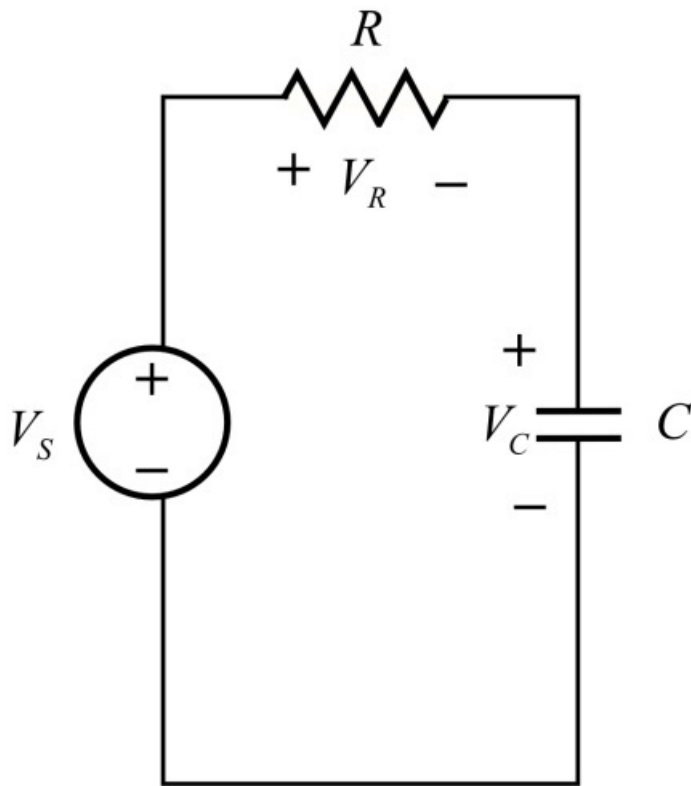


Figure 1

Step 2 of 2

Consider V_S is the voltage across the supply, V_R is the voltage across the resistor and V_C is the voltage across the capacitor.

Determine the magnitude of the supply voltage.

Write the relation between V_R , V_C , V_S as,

$$V_S^2 = |V_R|^2 + |V_C|^2$$

$$V_S = \sqrt{|V_R|^2 + |V_C|^2}$$

Replace 12 for V_R , 5 for V_C in equation.

$$\begin{aligned} V_S &= \sqrt{(12)^2 + (5)^2} \\ &= \sqrt{144 + 25} \\ &= \sqrt{169} \\ &= 13 \text{ V} \end{aligned}$$

Therefore, the magnitude of the supply voltage is 13 V.

Therefore, the correct option is C.